

Generation Mix Optimization Under Competing Objectives

A Parametric Framework for Climate and Resilience Tradeoffs

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1. The Problem

1.1 A Demand Surge Without Precedent

North America is experiencing the fastest growth in baseload electricity demand since electrification. The proximate cause is the hyperscale data center — a facility drawing 100 to 500 megawatts of continuous load from the bulk transmission system, operating at high utilization around the clock, every day of the year. These facilities are not intermittent loads with daily and seasonal variation. They are permanent additions to the baseload demand curve, indistinguishable in their load character from a small city that never sleeps.

The scale is not marginal. NERC’s current load growth projections identify data center demand as the dominant driver of new load across multiple North American interconnections. ERCOT alone is processing interconnection queue requests from data center campuses representing tens of gigawatts of new demand. The broader U.S. picture is consistent: every major transmission region is managing a queue of hyperscale load requests that outpaces the available generation in the interconnection queue ahead of it.

The investment community has recognized the opportunity. Venture capital, private equity, and infrastructure funds are committing capital to data center development at an unprecedented rate. The constraint they are now encountering is not land, fiber, or cooling — it is reliable, continuous, affordable electricity at the scale these facilities require. The generation question has moved from a background infrastructure assumption to the central investment variable.

1.2 Why the Generation Question Is Harder Than It Looks

The obvious answer — build more generation — immediately encounters a set of competing constraints that the investment community is not always equipped to evaluate simultaneously.

The first constraint is reliability. A hyperscale data center serving financial infrastructure, artificial intelligence inference, or defense-adjacent workloads operates under contractual and operational reliability requirements that approach one hundred percent availability. The Uptime Institute’s Tier IV classification — the industry standard for mission-critical data center infrastructure — specifies 99.995 percent availability, corresponding to approximately twenty-six minutes of allowable downtime per year. These are not aspirational targets; they are contractual commitments embedded in colocation agreements and tenant leases. The generation serving this load must match that reliability profile. A source that is available and producing ninety-two percent of the time is a fundamentally different investment proposition from one that is available ninety-seven percent of the time but producing only twenty-five percent of the time. The distinction between availability and productive output — the difference between a source being ready and a source actually generating — is one that standard investment analysis frequently conflates, with significant consequences for capital recovery projections.

The second constraint is carbon. The customers occupying hyperscale facilities — cloud providers, AI companies, financial institutions — are operating under public commitments to carbon neutrality or net-zero electricity consumption. A data center powered by natural gas at 490 grams of carbon dioxide per kilowatt-hour is incompatible with those commitments, regardless of its reliability profile. The generation investment must satisfy both the reliability requirement and the carbon requirement simultaneously. These two requirements are not always in tension, but the combinations that satisfy both are fewer than investment discussions typically acknowledge.

The third constraint is resilience. This is the least visible of the three in current market discussions, but it is increasingly consequential. Generation assets are physical infrastructure with physical vulnerabilities. The targeting of power generation and transmission infrastructure — demonstrated repeatedly in active conflict zones and increasingly a concern for critical infrastructure planners in stable regions — introduces a set of risks that standard reliability analysis does not capture. A generation source that is reliable and low-carbon but recoverable only over an eighteen-to-twenty-four-month timeline following physical damage to its transmission interconnection presents a risk profile that differs qualitatively from one that can be restored in weeks. Similarly, a source dependent on continuous fuel delivery through a pipeline lateral that can be interrupted presents a different risk profile from one carrying months of fuel on site.

The investment community currently has no unified analytical framework for evaluating generation options across all three constraints simultaneously. Levelized cost of energy captures the economic dimension imperfectly and omits the reliability, carbon, and resilience dimensions almost entirely. Carbon intensity analyses address the climate constraint but ignore reliability and resilience. Reliability metrics address uptime but do not capture the distinction between availability and productive output, and do not address resilience under physical threat conditions. The result is that new-build investment decisions are made sequentially against individual constraints rather than simultaneously against the full constraint set — a process that systematically produces suboptimal outcomes.

1.3 The Purpose of This Paper

This paper develops a parametric framework for evaluating generation sources across all relevant objectives simultaneously. The framework builds on prior work by the author, establishing a parametric model of the bulk electric system and analyzing generation characteristics under conditions of physical duress. It extends that foundation in two directions: first, by formalizing the optimization structure that was implicit in the earlier work; second, by adding the resilience terms that the earlier climate-focused analysis did not require.

The framework is expressed as a multivariate function of eight parameters, each capturing a dimension of generation performance that investment, regulatory, or operational decisions depend on. The optimization of this function under competing objectives — minimize carbon

intensity, maximize capital recovery, maximize resilience — produces a Pareto frontier whose shape carries the analytical finding: the set of generation sources that perform well across all three objectives simultaneously is small, and the investment implications of that finding are significant for the data center demand surge now underway.

The mathematics in this paper is conceptual rather than fully solved. The optimization problem does not have a closed-form analytical solution for realistic grid configurations — numerical methods are required for site-specific analysis — and the value of the framework lies in the structure it imposes on the problem and the directional results it produces, not in numerical precision. The parameter values used throughout are analytically estimated from engineering first principles and published data sources where available, and stated as such where they are not.

The paper is organized as follows. Section 2 develops the parametric model and establishes the values table that serves as the empirical foundation. Section 3 derives the climate optimization and its implications for the generation mix. Section 4 applies the framework to the new-build investment decision, with particular attention to the data center demand surge. Section 5 extends the framework to include resilience terms and develops the Pareto frontier between climate and resilience objectives. Section 6 states the implications for investment, regulation, and policy. Section 7 identifies forward work.

2. The Parametric Model

2.1 The Bulk Electric System Framework

Electricity generation in North America is organized around four functional roles. Generation companies — GenCos — produce electricity from primary energy sources. Transmission companies — TransCos — move bulk power at high voltage across long distances. Distribution companies — DisCos — deliver power at lower voltage to end users. Control companies — CntlCos — balance generation and load in real time across the interconnection.

This paper is concerned with the GenCo function exclusively. TransCo, DisCo, and CntlCo each introduce their own vulnerability and reliability characteristics; they are outside the scope of the present analysis. The boundary condition at the GenCo output is the high-voltage terminal of the generator step-up transformer — the point at which generated power enters the transmission system.

Within the GenCo function, two broad classes of generation technology must be distinguished because their physical coupling to the grid differs in ways that matter for every objective in the optimization.

Synchronous generation — nuclear, coal, natural gas turbines, large hydro, and concentrating solar — produces electricity through a rotating synchronous generator directly coupled to the grid at the grid's electrical frequency. The rotating mass of the generator and its prime mover

stores kinetic energy that is exchanged with the grid automatically and instantaneously in response to frequency deviations. This exchange — called inertial response — is a physical property of the machine, not a control system function.

Inverter-based resources — utility-scale wind, utility-scale solar photovoltaic, and battery energy storage systems — produce electricity through power electronics that synthesize output current at commanded power and frequency. They do not exchange kinetic energy with the grid through rotating mass. Their coupling to the grid is through controls, not physics.

This distinction is not a technical detail. It is the variable that connects generation mix composition to grid frequency stability, and it is the reason that two portfolios with identical nameplate capacity and identical carbon intensity can have materially different grid stability and resilience profiles.

2.2 The Extended Generation Source Function

Eight parameters characterize a generation source. The source function is expressed as:

$$ES = f(CF, AF, LCOE, gCO_2/kWh, H, FID, RT_HVLLC, D(t))$$

CF — Capacity Factor is the ratio of actual energy produced over a period to the maximum possible energy production at nameplate output over the same period. CF is dimensionless, bounded on $[0,1]$. It measures productive utilization. A source with $CF = 0.92$ produces ninety-two percent of its theoretical maximum output averaged over the year. CF is a property of both the technology and the dispatch environment — a source may have a high technical CF but a lower realized CF if economic dispatch suppresses its output.

AF — Availability Factor is the proportion of time during which a source is mechanically capable of producing power, regardless of whether it is dispatched. AF is dimensionless, bounded on $[0,1]$. It measures mechanical readiness. The Uptime Institute Tier IV standard — the benchmark for mission-critical data center infrastructure — specifies 99.995 percent availability as the contractual floor for colocation facilities, establishing the reliability target that generation serving this load class must approach. The distinction between CF and AF is operationally significant: a source can have high AF and low CF if it is ready to produce but not dispatched, or low AF and moderate CF if it is frequently dispatched but also frequently offline for maintenance. The product $CF \times AF$ approximates expected utilization and is the relevant metric for capital recovery analysis.

LCOE — Levelized Cost of Energy is the present value of all costs over the plant lifetime divided by the present value of all energy produced, expressed in dollars per kilowatt-hour. LCOE is the standard economic comparison metric for generation options. It includes capital cost, financing cost, fuel cost, operations and maintenance, and decommissioning, spread over the plant's productive lifetime. LCOE alone is an insufficient basis for investment decisions because it does not capture the value of the energy produced — a source with low LCOE but low $CF \times AF$ may recover capital more slowly than a source with higher LCOE but higher $CF \times AF$.

gCO₂/kWh — Lifecycle Carbon Intensity measures the total greenhouse gas emissions associated with a generation source over its full lifecycle, normalized to the energy produced, expressed in grams of carbon dioxide equivalent per kilowatt-hour. Lifecycle accounting includes upstream emissions from fuel extraction and processing, manufacturing emissions from plant construction, operational emissions from fuel combustion and auxiliary systems, and downstream emissions from decommissioning. Operational emissions dominate for combustion sources. Construction and manufacturing emissions dominate for nuclear and renewables.

H — Synchronous Inertia Constant is the ratio of the kinetic energy stored in the rotating mass of a synchronous generator and its prime mover to the generator's rated power output, expressed in seconds. H quantifies how much energy a synchronous machine exchanges with the grid during a frequency disturbance before governor response or other control actions take effect. H is defined only for synchronous generation; inverter-based resources have $H = 0$ in standard grid-following configurations. Synthetic inertia emulation through grid-forming inverter control can provide an effective $H > 0$ for inverter-based resources, but this is not standard in current deployments and is noted as a variant condition where applicable. A supply chain risk dimension attaches to synthetic inertia that does not attach to physical inertia: the inverter devices through which synthetic inertia would be provided are manufactured predominantly by Chinese firms — Huawei and Sungrow alone hold approximately 40 to 50 percent of the global utility-scale inverter market share — and the control algorithms implementing synthetic inertia are proprietary firmware running on those devices. The reliability of synthetic inertia as a grid stability mechanism is therefore not a purely technical question. It is addressed further in Section 5.5.

FID — Fuel Independence Duration is the number of days a generation source can continue operating at full output under complete denial of external fuel supply, using only the energy stored or available on site at the time of supply interruption. FID is expressed in days. For sources whose fuel cannot be denied by physical interdiction — wind, solar, and run-of-river hydro — FID is treated as effectively unbounded for military action analysis purposes, with the caveat that the generating equipment itself can be destroyed. For sources dependent on continuous fuel delivery — natural gas in particular — FID is determined by pipeline linepack and any on-site storage. FID is an original analytical construct of this paper; no standard literature definition exists.

RT_HVLLC — Recovery Timeline for High-Value Long-Lead Components is the time required to restore a generation facility to service following destruction of its most difficult-to-replace major component, expressed in months. The binding constraint for all transmission-connected generation is the high-voltage main power transformer: a custom-manufactured component with typical lead times of twelve to twenty-four months under normal commercial conditions and no meaningful strategic reserve in North America. RT_HVLLC represents the maximum duration of forced outage following a precision physical attack on the facility's most vulnerable single component. RT_HVLLC is an original analytical construct of this paper.

D(t) — Demand Signal is the time-varying electricity demand the generation portfolio must serve, expressed in kilowatt-hours as a function of time. For data center load specifically, D(t) is approximately constant — high utilization around the clock with minimal daily or seasonal variation. This load character is distinct from residential and commercial load profiles and has significant implications for the generation mix required to serve it reliably.

2.3 Parameter Values by Source

Table 1 presents analytically estimated parameter values for eight generation source types relevant to the North American bulk electric system. Values are drawn from published sources where available — EIA, NERC, IPCC, IEEE — and from engineering first principles where no published standard exists. The latter are identified explicitly.

Values represent fleet averages or representative mid-range estimates unless otherwise noted. Site-specific values will vary; the table is intended to support comparative analysis across source types, not to substitute for site-specific engineering assessment.

Table 1: Generation Source Parameters

Parameter	Nuclear	Coal	CCGT	Simple-Cycle Gas	Large Hydro	Utility Wind	Utility Solar PV	BESS
CF	0.92	0.54	0.57	0.15	0.42	0.35	0.25	0.15
AF	0.90	0.78	0.85	0.92	0.88	0.97	0.99	0.95
CF × AF	0.83	0.42	0.48	0.14	0.37	0.34	0.25	0.14
LCOE (\$/MWh)	89	65	39	58	50	33	24	140*
gCO ₂ /kWh	12	820	490	650	24	11	48	†
H (seconds)	6.0	5.0	4.5	3.5	4.0	0	0	0
FID (days)	270	30	0.5	0.5	∞	∞	∞	0.17
RT-HVLLC (months)	18–24	18–24	18–24	12–18	24–36	6–12	3–6	6–12

Sources: CF and AF — EIA Electric Power Annual 2024, NERC 2025 Long-Term Reliability Assessment. LCOE — Lazard Levelized Cost of Energy Analysis v17. gCO₂/kWh — IPCC AR6 lifecycle emissions ranges, median values. H — IEEE Std 1110, analytically estimated by the author from machine class data. FID — analytically estimated by the author from fuel storage and pipeline engineering first principles. RT_HVLLC — analytically estimated by the author from equipment manufacturing lead time data and infrastructure targeting open-source analysis.

*BESS LCOE reflects storage cost only; charging source cost is additional.

†BESS gCO₂/kWh is determined entirely by the carbon intensity of the charging source and has no intrinsic value.

2.4 What the Table Immediately Reveals

Three observations emerge from Table 1 before any optimization is performed.

First, the $CF \times AF$ product — the expected utilization rate — separates the generation landscape into two groups with a wide gap between them. Nuclear at 0.83 and CCGT at 0.48 occupy the upper range. Utility wind at 0.34, utility solar at 0.25, and simple-cycle gas and BESS at 0.14 occupy the lower range. This gap has direct capital recovery implications: at a given electricity price, a source with $CF \times AF = 0.83$ recovers its capital investment in roughly one-third the time of a source with $CF \times AF = 0.25$. For the data center demand surge — where load is continuous, and capital must be recovered over a plant lifetime — this gap is not a technical detail. It is the central investment variable.

Second, the H column divides the table cleanly at zero. All synchronous sources carry positive inertia constants in the range of 3.5 to 6.0 seconds. All inverter-based sources carry zero inertia in standard configurations. As the generation mix shifts from synchronous to inverter-based resources, the system inertia available to arrest frequency deviations following a generation loss event declines proportionally. This decline is not visible in CF, AF, LCOE, or gCO₂/kWh — it is a hidden cost of decarbonization strategies that do not account for grid stability explicitly.

Third, the FID column reveals a vulnerability structure that no existing generation planning framework captures. Natural gas — the fuel source currently serving the majority of new data center load due to its fast permitting and construction timeline — has an FID of 0.5 days. Under conditions of fuel supply interruption, a gas-fired plant serving a hyperscale data center goes dark within hours. Coal has thirty days. Nuclear has two hundred and seventy days. The sources with unbounded FID — wind, solar, and hydro — are immune to fuel supply denial but subject to equipment destruction, the consequences of which are captured in the RT_HVLLC column.

These three observations frame the optimization problems developed in Sections 3 through 5.

3. The Climate Optimization

3.1 Framing the Problem

The climate objective is to minimize the carbon intensity of the electricity delivered to the load. For a portfolio of generation sources, carbon intensity is not the carbon intensity of any single source — it is a weighted average across all sources in the mix, weighted by their contribution to actual delivered energy. A source with very low carbon intensity but very low $CF \times AF$ contributes little to the weighted average in either direction. A source with high carbon intensity and high $CF \times AF$ dominates the portfolio carbon outcome.

This observation immediately suggests the correct optimization structure. The decision variable is not which sources to include — it is how much of each source to allocate. Define the portfolio allocation as a vector:

$$\mathbf{x} = (x_1, x_2, \dots, x_n)$$

where x_i is the fraction of total installed capacity allocated to source i , and the allocations sum to one:

$$\sum x_i = 1$$

The portfolio carbon intensity is then the weighted average of each source's lifecycle carbon intensity, weighted by its expected productive output:

$$G(\mathbf{x}) = \sum (g_i \times x_i \times CF_i \times AF_i) / \sum (x_i \times CF_i \times AF_i)$$

where g_i is the lifecycle carbon intensity of source i in gCO_2/kWh , CF_i is its capacity factor, and AF_i is its availability factor.

The denominator — the sum of productive output contributions — is the normalization that makes $G(\mathbf{x})$ a true intensity rather than a raw emissions sum. This normalization matters: adding a low-carbon source with low $CF \times AF$ improves $G(\mathbf{x})$ less than adding the same nameplate capacity of a low-carbon source with high $CF \times AF$, because the former contributes less productive output to the denominator.

3.2 The Demand Constraint

The portfolio must meet demand. For general load, the demand constraint is time-varying — the portfolio must serve $D(t)$ at every hour of the year, accounting for the coincidence of load with source availability. For data center load specifically, $D(t) \approx D$ — a constant. The data center draws its contracted load continuously, without the daily and seasonal variation that characterizes residential and commercial demand.

The demand constraint for a constant load is:

$$\sum (x_i \times CF_i \times AF_i \times C_{\text{total}}) \geq D$$

where C_{total} is the total installed portfolio capacity, this says that the expected productive output of the portfolio — the capacity-weighted sum of $CF \times AF$ products — must be sufficient to meet continuous demand D .

This constraint is where the $CF \times AF$ product from Table 1 becomes the binding variable. A portfolio composed entirely of utility solar ($CF \times AF = 0.25$) must do two things simultaneously to satisfy the same constant demand as a portfolio composed entirely of nuclear ($CF \times AF = 0.83$). First, it must install roughly four times the nameplate capacity to generate sufficient total energy over the producing hours. Second, it must store the surplus energy generated during those producing hours so that it is available during the hours when the source produces nothing. These two requirements are not alternatives — both are necessary, and both represent capital

investment that does not appear in the LCOE figure for the solar source alone. The nameplate capacity multiplier determines how much generation infrastructure must be built; the storage requirement determines how much storage infrastructure must be built alongside it. Omitting either understates the true capital requirement for continuous load service from a variable source.

3.3 The Lagrangian and First-Order Conditions

The constrained optimization is expressed as a Lagrangian — a standard technique for finding the minimum of an objective function subject to an equality constraint. The Lagrangian combines the objective function $G(x)$ with the demand constraint, weighted by a multiplier λ that represents the shadow price of meeting demand with low-carbon generation:

$$L(x, \lambda) = G(x) + \lambda[D - \sum(x_i \times CF_i \times AF_i \times C_total)]$$

The minimum of $G(x)$ subject to the demand constraint occurs where the partial derivatives of L with respect to each x_i equal zero. Beginning with the full derivative of the Lagrangian with respect to x_i :

$$\partial L / \partial x_i = \partial G / \partial x_i + \lambda \times \partial / \partial x_i [D - \sum(x_i \times CF_i \times AF_i \times C_total)] = 0$$

The demand target D is a constant with respect to x_i , so its derivative is zero. The derivative of the sum term with respect to x_i picks out only the i -th term, since all other terms in the sum do not depend on x_i :

$$\partial L / \partial x_i = \partial G / \partial x_i - \lambda \times CF_i \times AF_i \times C_total = 0$$

Setting this equal to zero and rearranging — moving the second term to the right side — gives the first-order condition for source i :

$$\partial L / \partial x_i = \partial G / \partial x_i - \lambda \times CF_i \times AF_i \times C_total = 0$$

Rearranging:

$$\partial G / \partial x_i = \lambda \times CF_i \times AF_i \times C_total$$

This condition has a direct physical interpretation that does not require solving the system algebraically. At the carbon-optimal portfolio, the marginal reduction in portfolio carbon intensity from increasing the allocation of source i must equal the shadow price of the demand constraint times the productive output contribution of source i .

Reading this condition across the sources in Table 1 produces the key structural result. Sources with high $CF_i \times AF_i$ appear with large weight on the right side — they contribute substantially to meeting the demand constraint. For such a source to satisfy the first-order condition, its marginal carbon reduction $\partial G / \partial x_i$ must also be large — meaning the source must have low carbon intensity g_i relative to the current portfolio average. Sources with high $CF \times AF$ and low g_i satisfy this condition strongly. Sources with high $CF \times AF$ and high g_i violate it — their demand contribution is large, but their carbon contribution worsens the portfolio outcome.

From Table 1, reading across $CF \times AF$ and gCO_2/kWh simultaneously: nuclear ($CF \times AF = 0.83$, $gCO_2/kWh = 12$) strongly satisfies the first-order condition. Large hydro ($CF \times AF = 0.37$, $gCO_2/kWh = 24$) satisfies it well within its resource constraint. Utility wind ($CF \times AF = 0.34$, $gCO_2/kWh = 11$) satisfies it well, but large nameplate capacity is required to contribute meaningfully to demand. CCGT ($CF \times AF = 0.48$, $gCO_2/kWh = 490$) strongly violates the first-order condition — its productive output contribution is valuable, but its carbon cost is disqualifying in a carbon-minimizing portfolio.

3.4 The AF Constraint and the Baseload Gap

The first-order conditions identify the preferred allocation direction but do not by themselves reveal why renewable sources cannot satisfy the demand constraint alone without storage. That result emerges from the demand constraint directly.

For constant data center demand D , the portfolio must maintain $\sum(x_i \times CF_i \times AF_i \times C_total) \geq D$ at all times. The CF values in Table 1 are annual averages — they do not capture the hours during which a given source produces nothing at all. Solar produces zero output for approximately half of every twenty-four-hour period and substantially reduced output during low-angle winter sunlight. Wind produces zero output during calm periods that can extend for days across entire regional geographies.

During these zero-production periods, the demand constraint requires that other sources in the portfolio fill the gap. If the portfolio contains only solar and wind, the demand constraint requires either that their combined installed capacity is large enough that even during simultaneous low-production periods the residual output meets D — which implies massive overcapacity at peak production — or that storage of sufficient duration is included to bridge the gap.

The storage requirement is not a planning assumption. It is a mathematical consequence of the demand constraint applied to sources with variable CF . A four-hour BESS bridges a four-hour gap. Bridging the multi-day calm periods that define wind's worst-case CF requires storage measured in days to weeks of portfolio output. At the current BESS LCOE of \$140/MWh, the capital cost of that storage transforms the apparent LCOE advantage of wind and solar into a significant cost premium relative to continuously available sources.

This result — that the demand constraint imposes a storage requirement whose cost is not captured in the LCOE of the variable source alone — is one of the most important and most frequently omitted findings in generation mix analysis. The parametric framework makes it explicit because the demand constraint is stated correctly as a function of $CF \times AF$ rather than nameplate capacity.

3.5 The Sensitivity Result

The most policy-relevant output of the climate optimization is not the optimal portfolio itself — it is the sensitivity of the optimal carbon intensity to constraints on individual source allocations. Specifically: how much does the achievable $G(x)$ worsen as the maximum nuclear allocation x_{nuclear} is progressively reduced?

This sensitivity is expressed as $\partial G^*/\partial x_{\text{nuclear_max}}$ — the rate of change of the minimum achievable portfolio carbon intensity as the nuclear ceiling is lowered. The sign is unambiguous from the first-order conditions: $\partial G^*/\partial x_{\text{nuclear_max}} < 0$, meaning that reducing the nuclear ceiling always worsens the achievable carbon outcome. The magnitude determines how costly the constraint is.

Because nuclear has the highest $CF \times AF$ in the table and the second-lowest carbon intensity, reducing its allocation forces the optimization to substitute sources that are either lower in $CF \times AF$ — requiring more nameplate capacity and storage to maintain the demand constraint — or higher in carbon intensity — directly worsening $G(x)$ — or both. No other source in Table 1 provides nuclear’s combination of high $CF \times AF$ and low gCO_2/kWh .

The practical implication is stated directly: a portfolio optimization that excludes nuclear on non-technical grounds accepts a carbon penalty that any combination of the remaining sources cannot recover without either accepting higher emissions from gas or accepting the capital cost of sufficient storage to firm variable renewables to a baseload profile. For the data center demand surge, where the load is continuous, and the carbon commitment is binding, that penalty is not marginal. It is structural. A numerical illustration of this sensitivity result is provided in the Technical Appendix.

4. The New-Build Investment Framework

4.1 Why LCOE Misleads the Investment Decision

The investment community’s reliance on LCOE as the primary generation comparison metric is not irrational — LCOE is a well-defined, consistently calculated figure that enables rapid comparison across technologies. The problem is not that LCOE is wrong. The problem is that it answers a question the data center investor is not asking.

The data center investor is asking: what is the cost per unit of continuously delivered, low-carbon, physically resilient electricity over a twenty-year contract horizon? LCOE answers a different question: what is the average cost per unit of electricity produced over a plant’s lifetime under normal operating conditions? The gap between these two questions is where investment errors accumulate.

The parametric framework closes this gap by providing a structured method for moving from LCOE to the metric the investor actually needs. The progression is three steps. Step one: Adjust for the continuous delivery requirement by computing the Firmed LCOE. Step two: adjust for

capital recovery timeline as a function of $CF \times AF$. Step three: adjust for resilience-related downside risk by incorporating FID and RT_HVLLC into the risk-adjusted return calculation.

The Resilience-Adjusted Delivered Cost (RADC)

This paper proposes a metric suited to the data center investment decision: the Resilience-Adjusted Delivered Cost, or RADC. RADC is defined as the cost per unit of continuously delivered, low-carbon, physically resilient electricity over the contract horizon. It incorporates three adjustments that LCOE does not make: first, the cost of firming variable sources to continuous output, so that the comparison is made on delivered energy rather than nameplate production; second, the carbon intensity of the delivered energy measured against the operator's net-zero commitment, so that sources incompatible with that commitment are correctly excluded from the feasible set; and third, the resilience-adjusted downside risk captured by fuel independence duration and recovery timeline, so that the probability and consequence of forced outage under supply denial or physical damage are reflected in the cost comparison. RADC is the metric that the data center investment decision requires. LCOE is not.

4.2 Capital Recovery as a Function of $CF \times AF$

The capital recovery logic developed here formalizes a construct well established in utility finance: the Allowance for Funds Used During Construction, or AFUDC. AFUDC is the accounting mechanism by which the carrying cost of capital during the construction period is capitalized into a project's rate base and recovered only once the plant enters commercial operation and begins generating revenue. The AFUDC construct makes explicit what every project finance model already assumes: capital does not earn its return until the asset is producing. The productive utilization rate — $CF \times AF$ — is therefore the binding variable in the capital recovery timeline, not the installed cost alone.

Capital invested during construction — including the financing cost carried during the construction period — begins recovering only when the plant enters commercial operation and begins generating revenue. The revenue stream is proportional to energy delivered, which is proportional to $CF \times AF$.

For a plant with installed capacity C , electricity price P , and annual fixed costs F , the annual net revenue available for capital recovery is approximately:

$$R = (CF \times AF \times C \times P) - F$$

The capital recovery timeline T is the total capital investment K divided by the annual net revenue R :

$$T = K / R = K / [(CF \times AF \times C \times P) - F]$$

The partial derivative of T with respect to $CF \times AF$ is:

$$\partial T / \partial (CF \times AF) = -K \times C \times P / [(CF \times AF \times C \times P) - F]^2$$

The sign is negative throughout — increasing $CF \times AF$ always shortens the capital recovery timeline, for any values of K , C , P , and F . The magnitude grows as $CF \times AF$ falls — meaning the sensitivity of capital recovery to productive utilization is greatest precisely for the sources with the lowest $CF \times AF$. Small improvements in $CF \times AF$ at the low end of Table 1 produce larger reductions in recovery timeline than equivalent improvements at the high end.

4.3 The True Cost of Firming Variable Generation

For a data center load profile — constant, continuous, approximately equal to D at all hours — the generation portfolio must deliver D reliably at every hour. A variable source with $CF \times AF$ below the demand threshold requires firming: either storage sufficient to bridge zero-production periods, or backup generation that operates during those periods.

The true capital cost of serving a continuous load from a variable source is not the LCOE of the variable source alone. It is the LCOE of the variable source plus the amortized cost of the firming infrastructure required to make its output continuous. Define this as the **Firmed LCOE**, which is itself a component of the broader RADC metric introduced in Section 4.1:

$$\text{LCOE}_{\text{firmed}} = \text{LCOE}_{\text{source}} + \text{LCOE}_{\text{storage}} \times (1 - CF \times AF_{\text{source}})^1$$

From Table 1 and the BESS LCOE of \$140/MWh:

$$\text{Utility solar firmed: } \$24 + \$140 \times (1 - 0.25) = \$24 + \$105 = \$129/\text{MWh}$$

$$\text{Utility wind firmed: } \$33 + \$140 \times (1 - 0.35) = \$33 + \$91 = \$124/\text{MWh}$$

$$\text{Nuclear: } \$89 + \$140 \times (1 - 0.92) = \$89 + \$11 = \$100/\text{MWh}$$

Two results are immediately apparent. First, the apparent LCOE advantage of solar and wind over nuclear inverts when firming cost is included. Firmed solar and firmed wind are more expensive than nuclear per unit of continuously delivered energy. Second, CCGT and nuclear arrive at nearly identical firmed LCOE — but with radically different carbon intensity (490 versus 12 gCO₂/kWh), resilience profile, and long-term fuel cost exposure.

¹The Firmed LCOE formula requires careful interpretation for solar and wind separately. For solar, four-hour BESS storage does not constitute a complete firming solution — it is a categorical inadequacy, not merely a quantitative underestimate. Solar produces no output for a minimum of ten to twelve hours every night regardless of season, and longer at higher latitudes in winter. A four-hour BESS bridges a partial evening gap; it does not bridge an overnight gap. A data center served by solar with four-hour storage goes dark well before sunrise. Firming solar to true continuous output requires a minimum of twelve to sixteen hours of storage for a mid-latitude site, substantially more for northern latitudes or winter operation. For wind, the inadequacy is quantitative rather than categorical: four-hour storage handles short-duration lulls but does not cover multi-day calm periods, which can persist for three or more consecutive days across wide geographic areas and which represent wind's worst-case reliability condition. Bridging multi-day calm events requires storage measured in days of portfolio output, not hours. Both firmed LCOE estimates in Section 4.3 are therefore conservative in the direction that favors variable sources, but for different reasons: the solar estimate understates a structural inadequacy, while the wind estimate understates a duration inadequacy. Actual firmed costs for continuous load profiles will be materially higher than stated for both sources.

4.4 The Data Center Demand Surge as Investment Context

The hyperscale data center represents a new class of electricity customer whose load characteristics align almost perfectly with the operational profile of baseload synchronous generation and almost imperfectly with the operational profile of variable renewables without storage.

Three load characteristics define the alignment. Continuity: a hyperscale campus draws its contracted load at all hours, requiring generation that is flat or gap-filled by storage at additional capital cost. Scale: individual campuses in the current interconnection queue are requesting 100 to 500 megawatts of firm capacity. At a 500 MW campus, the difference between firmed solar at \$129/MWh and nuclear at \$100/MWh represents approximately \$127 million per year in additional electricity cost — a material variable over a twenty-year power purchase agreement. Carbon commitment: the hyperscale operators are under public net-zero electricity commitments that CCGT at 490 gCO₂/kWh cannot satisfy, but nuclear at 12 gCO₂/kWh can.

4.5 The New-Build Comparison

For an investor evaluating new-build generation to serve a hyperscale data center load under a twenty-year power purchase agreement, the relevant comparison is across five metrics simultaneously: Firmed LCOE, capital recovery timeline, gCO₂/kWh, FID, and RT_HVLLC. Table 2 presents this comparison for the four most plausible new-build options in the current North American market.

Table 2: New-Build Comparison for Continuous Load Service

Metric	Nuclear (SMR)	CCGT	Wind + Storage	Solar + Storage
CF × AF	0.83	0.48	0.34	0.25
Firmed LCOE (\$/MWh)	~\$100	~\$99	~\$124	~\$129
Capital Recovery	Fastest among low-carbon	Fast	Slow	Slowest
gCO ₂ /kWh	12	490	11	48
FID (days)	270	0.5	∞*	∞*
RT-HVLLC (months)	18–24	18–24	6–12	3–6

*Wind and solar FID is unbounded for fuel supply denial; equipment destruction vulnerability is captured in RT_HVLLC.

The table makes visible what no single-metric analysis reveals. CCGT and nuclear arrive at nearly identical firm LCOE — but CCGT fails the carbon test decisively and fails the FID test within hours of any fuel supply disruption. Wind and solar satisfy the carbon test and the FID

test — but fail the firm LCOE test and the capital recovery test for continuous load. Nuclear is the only option that satisfies all five metrics simultaneously within acceptable ranges.

This is not an assertion about nuclear power policy. It is a mathematical consequence of the parameter values in Table 1 applied to the specific load characteristics of a hyperscale data center. A different load profile would produce a different comparison. The framework is neutral; the result follows from the inputs.

4.6 The CapEx Framing for the Investment Community

Nuclear carries the highest upfront capital cost of any generation technology — current SMR estimates range from \$5,000 to \$10,000 per kilowatt of installed capacity, compared to \$1,000 to \$1,500 per kilowatt for CCGT and \$1,200 to \$1,800 per kilowatt for utility solar. This capital cost differential is real and cannot be dismissed.

What the parametric framework reveals is that the relevant comparison is not \$/kW installed. It is \$/kW of continuously delivered, low-carbon, resilient output. Three adjustments are required. First, divide by $CF \times AF$ to convert installed capacity cost to cost per unit of reliable output: nuclear at \$7,500/kW installed becomes \$9,036 per kW of reliable continuous output; solar at \$1,500/kW nameplate requires three to four times the nameplate capacity plus storage to produce equivalent continuous output. Second, extend the comparison over plant lifetime: nuclear plants are licensed for sixty to eighty years versus twenty-five to thirty for solar and wind, changing the present value of the investment substantially at any reasonable discount rate. Third, account for fuel price exposure: nuclear's fuel cost is a small, stable fraction of its LCOE; CCGT's fuel cost is the dominant LCOE component and carries twenty-year natural gas price exposure.

When these three adjustments are applied, the CapEx case for nuclear in the data center context is considerably stronger than the headline \$/kW installed figure suggests. The framework does not eliminate the upfront capital challenge — that is real and requires appropriate financing structures, including the federal loan guarantees and production tax credits that the current policy environment provides. But it reframes the challenge correctly: the question is not whether nuclear is expensive, but whether it is expensive relative to the alternatives when all relevant costs and properties are counted.

5. The Resilience Optimization and the Pareto Frontier

5.1 Why Resilience Is a Separate Objective

The climate optimization in Section 3 and the investment framework in Section 4 together address two of the three constraints identified in Section 1: carbon intensity and capital recovery. The third constraint — resilience under conditions of physical threat and supply denial — requires a separate analytical treatment because it is not reducible to the first two.

Resilience in this context has a specific technical meaning. A generation source is resilient to the extent that it continues operating under conditions of external stress, recovers quickly when damaged, and imposes low consequence on the grid and on dependent loads when it fails. Three of the eight parameters in Table 1 capture this property: H, FID, and RT_HVLLC.

These three parameters were not part of the climate optimization because they do not appear in the carbon intensity function $G(x)$. A portfolio can be carbon-optimal while simultaneously being fragile: carrying low system inertia, short fuel independence, and long recovery timelines. Conversely, a portfolio can be highly resilient while carrying high carbon intensity. The relationship between climate optimality and resilience optimality is not one of alignment — it is one of partial conflict, and the nature of that conflict is what the Pareto frontier makes explicit.

5.2 The Resilience Function

Define the portfolio resilience as a composite function $R(x)$ that captures the three resilience dimensions simultaneously. A fully rigorous composite function requires weighting the three terms against each other — a weighting that involves judgments about the relative severity of inertia deficiency, fuel supply denial, and component destruction that are outside the scope of this paper. Instead, the resilience function is expressed as a vector:

$$R(x) = (H_system(x), FID_portfolio(x), RT_HVLLC_portfolio(x))$$

$H_system(x)$ is the capacity-weighted average synchronous inertia constant of the portfolio:

$$H_system = \sum(x_i \times H_i \times CF_i) / \sum(x_i \times CF_i)$$

The weighting by CF reflects the fact that inertia contribution depends on the unit being online and synchronized to the grid. $FID_portfolio(x)$ is the demand-weighted minimum fuel independence duration across all sources in the portfolio; the minimum operator is appropriate because the portfolio's fuel independence is constrained by its least-independent component at any given dispatch condition. $RT_HVLLC_portfolio(x)$ is the capacity-weighted average recovery timeline across all transmission-connected sources in the portfolio.

5.3 The Partial Derivatives of the Resilience Function

The directional relationships between portfolio allocation and each resilience component follow from the parameter values in Table 1. As with the climate optimization, the derivatives are stated and interpreted rather than solved.

$\partial H_system / \partial x_IBR < 0$ — increasing the allocation of any inverter-based resource reduces system inertia. The rate of reduction is proportional to the CF of the IBR being added.

$\partial H_system / \partial x_sync > 0$ — increasing the allocation of any synchronous source increases system inertia, proportional to that source's H value. Among the synchronous sources in Table 1, nuclear contributes most per unit of online generation ($H = 6.0$).

$\partial \text{FID_portfolio} / \partial x_{\text{nuclear}} > 0$ — increasing nuclear allocation improves portfolio fuel independence, because nuclear has the highest FID of any dispatchable source at 270 days.

$\partial \text{FID_portfolio} / \partial x_{\text{gas}} < 0$ — increasing gas allocation worsens portfolio fuel independence. Even a small gas allocation creates a fuel supply vulnerability that constrains the portfolio's effective FID to the dispatch conditions of the gas component.

$\partial \text{RT_HVLLC} / \partial x_{\text{hydro}} > 0$ — increasing large hydro allocation increases the portfolio's average recovery timeline, because large hydro has the longest RT_HVLLC in the table at 24–36 months. This is the result that distinguishes large hydro from its otherwise favorable profile.

5.4 The Pareto Frontier

The climate objective $G(x)$ and the resilience objective $R(x)$ define two optimization problems over the same allocation vector x . The Pareto frontier is the set of allocations where no further improvement in $G(x)$ can be achieved without worsening at least one component of $R(x)$, and no further improvement in $R(x)$ can be achieved without worsening $G(x)$.

The shape of the Pareto frontier is determined by which sources appear in both the climate-optimal and the resilience-optimal portfolios.

Nuclear is Pareto-dominant. It contributes the highest H among all sources, the longest FID among dispatchable sources, and the second-lowest carbon intensity in the table. Increasing nuclear allocation improves both $G(x)$ and all three components of $R(x)$ simultaneously. There is no allocation of nuclear that worsens either objective. The optimization pushes nuclear allocation to its maximum feasible level — constrained by permitting, siting, capital availability, and grid integration limits — before the Pareto frontier is encountered.

Large hydro is near-dominant but qualified. It contributes low carbon intensity, positive inertia, and unbounded FID — favorable on all climate and most resilience terms. The qualification is RT_HVLLC: at 24–36 months, large hydro has the longest recovery timeline in the table, worsening the RT_HVLLC component of $R(x)$.

Wind and solar define the tradeoff region. They contribute the lowest carbon intensity in the table — favorable for $G(x)$ — but $H = 0$ in standard configurations, worsening H_{system} . Adding wind or solar improves $G(x)$ while worsening H_{system} . This is the core climate-resilience tradeoff.

CCGT and coal occupy the dominated region. They contribute positive inertia — favorable for H_{system} — but high carbon intensity unfavorable for $G(x)$. And CCGT's FID of 0.5 days makes it the worst performer on fuel independence in the dispatchable category. No allocation of CCGT or coal improves both $G(x)$ and $R(x)$ simultaneously. They are Pareto-dominated by nuclear across both objectives.

5.5 The Hidden Cost of Inertia Deficiency

The H_{system} component of the resilience function carries an economic consequence not captured anywhere in standard generation planning or investment analysis. This consequence operates through the swing equation — the physical relationship that governs grid frequency response following a generation loss event:

$$df/dt = (f_0 / 2H_{\text{system}}) \times (\Delta P / S_{\text{rated}})$$

Where f_0 is nominal frequency, ΔP is the generation-load imbalance, and S_{rated} is system rated capacity. For a fixed loss event ΔP , halving H_{system} doubles the rate of frequency decline — the ROCOF. A faster ROCOF produces a deeper frequency nadir before governor response and load shedding arrest the decline.

If the nadir breaches the under-frequency load shedding threshold — typically 59.5 Hz in North American systems — automatic relays disconnect blocks of load to arrest the decline. Load shedding is, from the perspective of a data center operator, an outage. It is not a grid failure in the traditional sense, but its consequence for the disconnected load is identical to a supply interruption.

The economic implication is direct: a portfolio that optimizes $G(x)$ by maximizing wind and solar allocation — progressively reducing H_{system} toward zero — increases the probability that any given generation loss event will produce a frequency nadir that triggers load shedding affecting the data center. This probability is not currently priced into generation investment decisions. H_{system} is not a market product. The generator that reduces H_{system} by displacing synchronous generation does not pay for the increased outage risk it imposes on continuous load customers. This is a market externality in the technical economic sense: a cost imposed on others not reflected in the price signals facing the decision maker. Its regulatory implications are addressed briefly in Section 6.

5.6 The Supply Chain Risk Dimension of Synthetic Inertia

The analysis in Section 5.5 treats synthetic inertia from grid-forming inverters as a technically available but not yet standard capability. A further dimension requires explicit treatment: the supply chain and geopolitical risk profile of the devices through which synthetic inertia would be provided.

Physical inertia from synchronous generators is a property of rotating mass. It cannot be compromised through firmware, disabled through a software update, or manipulated through a remote command. It operates below the control system layer entirely. Synthetic inertia from grid-forming inverters has none of these properties. It is a control algorithm running on a digital signal processor in a power electronics device, dependent on proprietary firmware, and in principle accessible through whatever communication interfaces the manufacturer has implemented.

The supply chain concentration of utility-scale inverters in the North American market is severe. Chinese manufacturers — led by Huawei and Sungrow, with additional firms including

GNMAX, GOODWE, and Growatt — collectively dominate global utility inverter production. Chinese firms control approximately 80 to 85 percent of global solar panel manufacturing by volume, approximately 80 percent of polysilicon production, and approximately 95 percent of wafer production. Rare earth permanent magnet processing — critical for direct-drive wind generators — is approximately 85 to 90 percent Chinese-controlled at the processing stage. The inverter market concentration is therefore part of a broader pattern of Chinese dominance across the IBR supply chain that extends from raw materials through finished devices.

CISA and the Department of Energy have both raised specific concerns about Chinese-manufactured inverters deployed in the United States grid. Documented cases of undisclosed remote access functionality — communication capabilities not disclosed in product specifications — have been identified in inverters from Chinese manufacturers already installed in operational facilities. These findings establish that the risk is not hypothetical. The devices are in the grid, the undisclosed capabilities exist, and the regulatory framework for addressing them remains incomplete.

The grid stability implication is direct. A grid-forming inverter with synthetic inertia capability that is behaving adversarially — withdrawing inertia response at a moment of grid stress, or providing a destabilizing rather than stabilizing response — would be indistinguishable from normal inverter behavior until the moment of consequence. The attack surface is embedded in the generation fleet rather than external to it. This is a qualitatively different risk from the physical destruction addressed by RT_HVLLC: it does not require physical access, it does not leave forensic evidence at the facility, and it cannot be mitigated by physical hardening.

For the portfolio optimization developed in this paper, the SCRM dimension reinforces rather than complicates the directional results. Synthetic inertia from Chinese-manufactured inverters is not a reliable substitute for physical inertia from domestically sourced synchronous generation. The $H = 0$ treatment for IBRs in Table 1, and the associated partial derivative $\partial H_{\text{system}} / \partial x_{\text{IBR}} < 0$, correctly reflects the effective inertia contribution of IBRs in a supply-chain-aware assessment. The SCRM dimension is an additional analytical layer on top of the technical result, not a revision of it.

5.7 The Integrated Result

The three sections of the optimization — climate, investment, and resilience — converge on a single integrated finding. For the specific problem of serving continuous, large-scale, low-carbon, resilient electricity demand — which is precisely the problem the hyperscale data center surge has placed before the investment community — the generation source that satisfies all objectives simultaneously within acceptable ranges is nuclear. Not because of any assumption built into the framework, but because of the parameter values in Table 1 applied consistently across all three objective functions.

This finding does not resolve the capital cost challenge, the permitting timeline, or the social and political constraints on nuclear development. It does establish, on an analytical basis, what the cost of those constraints is: a portfolio that excludes nuclear must accept either higher carbon intensity, higher firm electricity cost, lower grid resilience, or some combination of all three. The framework quantifies the tradeoff. The policy judgment about whether that tradeoff is acceptable belongs to decision makers, not to the optimization.

6. Implications

6.1 Framing

The parametric framework developed in Sections 2 through 5 produces analytical results across three domains: the engineering optimization, the investment decision, and the policy environment. The engineering results are embedded in the mathematics of Sections 3 and 5. This section draws out the implications that flow from those results for the two domains where the gap between current practice and what the framework reveals is widest: investment decision-making and the regulatory policy framework governing generation adequacy and resilience.

The inertia externality — the unpriced grid stability cost imposed by progressive displacement of synchronous generation — is identified and analytically grounded in Section 5.5. The supply chain risk dimension of synthetic inertia — Chinese manufacturing concentration in utility inverters and the associated firmware and remote access risk — is addressed in Section 5.6. Both involve NERC standards development, market design, and critical infrastructure protection questions that are active areas of work in the reliability and security communities. This section does not repeat those treatments.

6.2 Investment Implications

6.2.1 The Metric Problem

The investment community's reliance on LCOE as the primary generation comparison metric is not irrational — LCOE is a well-defined, consistently calculated figure. The problem is that it answers a question the data center investor is not asking. The data center investor is asking: what is the cost per unit of continuously delivered, low-carbon, physically resilient electricity over a twenty-year contract horizon? The gap between that question and what LCOE answers is where investment errors accumulate.

The parametric framework closes this gap through a three-step progression. Step one: adjust for continuous delivery requirement by computing Firm LCOE as developed in Section 4.3. Step two: adjust for capital recovery timeline by computing $T = K / R$ as a function of $CF \times AF$, revealing how long invested capital is at risk before recovery. Step three: adjust for resilience-related downside risk by incorporating FID and RT_HVLLC into the risk-adjusted return

calculation — a source with FID = 0.5 days carries pipeline supply interruption as a force majeure risk that could void the power purchase agreement during exactly the conditions when the data center operator most needs power. These three adjustments require only that the parameters in Table 1 be applied to the investment calculation consistently.

6.2.2 The Portfolio Composition Signal

For investors assembling generation portfolios, the Pareto frontier result in Section 5.4 carries a portfolio construction implication. The efficient frontier of generation portfolios — minimizing carbon intensity for a given level of resilience, or maximizing resilience for a given carbon intensity — is concentrated in a small region defined by the Pareto-dominant and near-dominant sources.

Nuclear and large hydro dominate the efficient frontier. Wind and solar occupy the tradeoff region where carbon improvement comes at inertia cost. CCGT and coal are dominated across both objectives for long-term portfolio construction. For a portfolio investor seeking to serve the data center demand surge with a generation mix satisfying carbon commitments and resilience requirements simultaneously, the construction priority follows: maximize nuclear and hydro allocation first, within available resource and capital constraints; add wind and solar in the tradeoff region with explicit recognition of the H₂ system cost being incurred; use CCGT only as a transition-period instrument with a defined retirement schedule as preferred sources come online.

6.2.3 The Financing Structure Implication

The upfront capital cost of nuclear — the primary VC objection identified in Section 4.6 — is addressable within existing policy structures if the investment case is framed correctly. The current U.S. policy environment provides three financing instruments that directly address the nuclear CapEx challenge: Department of Energy loan guarantees under Title XVII of the Energy Policy Act, the production tax credit for advanced nuclear under the Inflation Reduction Act, and the investment tax credit available for new nuclear construction. Combined, these instruments can reduce the effective cost of capital for nuclear new-build by several hundred basis points.

The investment community has been slow to engage with these instruments for nuclear because the standard LCOE framing makes nuclear appear uncompetitive before financing structures are applied. The parametric framework inverts this: when the comparison is made on Firmed LCOE, capital recovery timeline, and resilience-adjusted return, nuclear enters the comparison as a competitive option to which financing optimization is then applied — rather than as an expensive option that requires subsidies to become viable. The analytical starting point changes the financing conversation.

6.3 Policy Implications

6.3.1 The Regulatory Gap in Generation Adequacy Standards

Current generation adequacy standards in North America — administered primarily through NERC reliability standards and regional transmission organization capacity market rules — measure generation adequacy in terms of installed capacity and planning reserve margins. The parametric framework identifies three specific regulatory gaps that current adequacy standards do not address.

The first gap is the CF $\times$ AF distinction. Planning reserve margins are calculated on nameplate capacity, not expected productive output. A reserve margin that appears adequate on a nameplate basis may be inadequate on a CF $\times$ AF basis if the reserve is disproportionately composed of low-CF $\times$ AF sources. For the data center demand surge — which adds continuous baseload load, not peak load — a nameplate-based adequacy standard systematically underestimates the generation requirement.

The second gap is FID. No current North American reliability standard addresses fuel independence duration as a generation adequacy metric. The result is that a generation fleet with arbitrarily short FID satisfies current adequacy standards while being structurally vulnerable to fuel supply interruption. The data center sector, which has no mandatory physical security or generation resilience standards of its own, is entirely dependent on grid-level adequacy standards that do not address this vulnerability.

The third gap is RT_HVLLC. The strategic reserve of large power transformers in North America is insufficient to support rapid recovery from coordinated physical attacks on transmission infrastructure. Current NERC standards address physical security of critical transmission assets under CIP-014 but do not establish recovery timeline requirements or strategic reserve obligations.

6.3.2 The Cross-Sector Dependency Problem

The data center sector illustrates a structural problem in the current critical infrastructure protection framework that extends beyond generation adequacy. Data centers are critical infrastructure — they underlie financial systems, communications networks, defense logistics, and emergency response — but they are not subject to mandatory physical security or generation resilience standards equivalent to those applied to nuclear facilities or bulk electric system assets.

The result is a dependency chain that current regulation does not address end-to-end. The bulk electric system is regulated for reliability and physical security through NERC standards and FERC oversight. The nuclear sector through NRC requirements. The natural gas sector has TSA security directives. The data center sector has no equivalent mandatory framework. This regulatory asymmetry means that investments in generation resilience can be undermined at the load end of the chain: a nuclear plant providing highly resilient generation to a hyperscale campus improves generation-side resilience, but if the data center itself has no mandatory

resilience standard, the system resilience gain may not translate into end-to-end service continuity.

6.3.3 The FID and RT_HVLLC Regulatory Pathway

The two resilience parameters that current regulation does not address — FID and RT_HVLLC — are analytically defined, can be calculated from available engineering data, and map onto existing regulatory frameworks in ways that do not require new statutory authority.

FID maps onto fuel supply reliability requirements that NERC already has authority to address through its reliability standards process. A minimum FID standard for generation serving critical load — expressed as a mandatory on-site fuel storage requirement or a contractual fuel supply diversity requirement — could be developed through the existing NERC Standards Authorization Request process without new legislation.

RT_HVLLC maps onto the physical security and critical equipment provisions of NERC CIP-014. Extending CIP-014 to include recovery timeline requirements — specifically, minimum strategic reserve obligations for long-lead transmission components — would require a standards revision but not new statutory authority. The Department of Energy has existing authority under the Federal Power Act to address strategic equipment reserves and has used that authority in related contexts. These regulatory pathways are identified here as forward work, not as fully developed proposals.

7. Conclusion and Forward Work

7.1 What This Paper Has Established

This paper has developed a parametric framework for evaluating electricity generation sources across the three objectives that matter simultaneously for the data center demand surge: carbon intensity, capital recovery, and physical resilience. The framework is built on eight parameters whose values for eight generation source types are established in Table 1 and whose interactions are analyzed through a structured optimization.

Four analytical results emerge from the framework. First, the Firmed LCOE and RADC result: the apparent cost advantage of variable renewable sources over nuclear inverts when the cost of firming those sources to continuous output is included, and the Resilience-Adjusted Delivered Cost — the metric that captures continuous delivery, carbon intensity, and physical resilience simultaneously — identifies nuclear as the most cost-effective option for the data center load profile. Second, the capital recovery result: the partial derivative of capital recovery timeline with respect to $CF \times AF$ is everywhere negative, and nuclear's $CF \times AF$ of 0.83 places it at the top of the utilization range, formalizing the AFUDC logic that productive utilization is the binding variable in capital recovery. Third, the Pareto dominance result: nuclear is the only generation source in the analysis that contributes favorably to both the climate objective and all

three components of the resilience objective simultaneously. Fourth, the regulatory gap identification: $CF \times AF$ as a continuous delivery metric, FID as a fuel independence metric, and RT_HVLLC as a recovery timeline metric are not addressed in any current North American reliability standard.

7.2 The Limitations of This Analysis

The framework developed in this paper is conceptual. The optimization problems are stated and their first-order conditions are interpreted, but they are not solved numerically. The parameter values in Table 1 are fleet averages and analytically estimated figures appropriate for comparative analysis across source types, not for site-specific investment decisions. Users of this framework for specific investment or planning applications should replace the Table 1 values with site-specific engineering assessments.

The resilience function $R(x)$ is expressed as a vector rather than a scalar composite because combining H, FID, and RT_HVLLC into a single weighted index requires value judgments about the relative severity of different failure modes that are outside the analytical scope of this paper. The Firmed LCOE formula applies a four-hour BESS firming assumption that understates true firming cost for continuous load profiles requiring multi-day storage. The paper does not address transmission costs, interconnection queue timelines, permitting durations, or workforce availability constraints, all of which affect the practical feasibility of new-build options.

7.3 Forward Work

The framework developed here is a foundation. Two extensions are identified as the primary forward research agenda.

The first extension is the physical constraint closure. The swing equation — which governs grid frequency response as a function of system inertia following a generation loss event — provides a physical differential equation that connects portfolio composition to grid stability consequence. Incorporating the swing equation as a binding physical constraint on the optimization closes the loop between the composition decision and the consequence of getting that decision wrong. This extension converts H_system from a resilience metric into an endogenous variable determined by the portfolio allocation, making the grid stability cost of inertia-reducing allocation decisions calculable rather than qualitative.

The second extension is the adversarial dual. The optimization developed in this paper asks: given a generation portfolio, what allocation minimizes carbon intensity subject to resilience and demand constraints? The adversarial dual asks the mirror question: given a generation portfolio, which unit should an adversary target to maximize grid consequence? The dual optimization identifies the highest-consequence target as a function of portfolio composition — the unit whose loss simultaneously maximizes the generation-load imbalance, minimizes the remaining H_system , and imposes the longest RT_HVLLC on the affected capacity. Solving the dual

produces a targeting priority ranking that is analytically derived rather than asserted, and whose dependence on portfolio composition makes it directly actionable for infrastructure protection planning.

These two extensions together would complete the analytical framework from the engineering optimization through the physical grid consequence to the adversarial targeting analysis. They are identified here as forward work because each requires a level of mathematical development beyond the conceptual scope of this paper, and each connects to active research agendas in the grid frequency protection and critical infrastructure security communities that warrant independent treatment.

7.4 Conclusion

The data center demand surge has placed before the investment community, the utility industry, and energy policy makers a generation adequacy problem of a scale and character that existing analytical frameworks were not designed to address. The demand is continuous, the carbon commitments are binding, the capital requirements are large, and the physical resilience requirements — while not yet reflected in any mandatory standard — are real and growing.

The parametric framework developed in this paper does not resolve those challenges. It provides a structured basis for analyzing them simultaneously rather than sequentially, and for making explicit the tradeoffs that single-objective analysis conceals. The finding that emerges — that nuclear is the only available source that satisfies all three objectives within acceptable ranges for this specific load profile — is not a policy conclusion. It is an analytical result. What policy makers, investors, and regulators do with that result involves judgments about capital markets, social preferences, and institutional constraints that are outside the scope of engineering optimization.

What is within scope is the insistence that those judgments be made with full visibility into what is being traded away. The parametric framework is a tool for that visibility. Its development into site-specific decision support, extended through the physical and adversarial analyses identified as forward work, is the research agenda this paper initiates.

Technical Appendix: Numerical Illustration of the Carbon Intensity Sensitivity Result

A.1 Purpose

Section 3.5 establishes that the partial derivative of minimum achievable portfolio carbon intensity with respect to the maximum nuclear allocation is everywhere negative — reducing the nuclear ceiling always worsens the achievable carbon outcome. This appendix provides a numerical illustration of that result using the parameter values from Table 1. No new analytical machinery is introduced. The illustration uses arithmetic applied to the weighted average carbon intensity formula developed in Section 3.1 under a simplified but representative portfolio assumption.

A.2 Setup

The illustration constructs a series of five portfolios, each satisfying the continuous demand constraint, with nuclear allocation progressively reduced from its carbon-optimal level toward zero. The remaining allocation in each portfolio is distributed across the next-best carbon performers — wind, solar, and large hydro — with gas-fired generation added only as required to satisfy the demand constraint when variable source allocation is insufficient to meet continuous load on a $CF \times AF$ basis.

The demand constraint is stated as a normalized continuous load requirement: a $CF \times AF$ portfolio product of 0.70. This is a representative planning figure for a mixed portfolio serving significant baseload load, deliberately conservative relative to a pure data center portfolio (which would require $CF \times AF \approx 0.90$), in order to show the sensitivity result across a range. Portfolio nameplate capacity is normalized to 1,000 MW. The carbon intensity formula is $G(x) = \Sigma(g_i \times x_i \times CF_i \times AF_i) / \Sigma(x_i \times CF_i \times AF_i)$. Parameter values are drawn directly from Table 1.

A.3 Portfolio Assumptions

In each portfolio, nuclear allocation is set at a specified ceiling. Large hydro is resource-constrained at a maximum of ten percent of portfolio capacity, reflecting geographic availability limits on new hydro in North America. Remaining allocation fills wind then solar. CCGT fills any remaining gap in the demand constraint as the minimum required to satisfy $CF \times AF \geq 0.70$ after all lower-carbon sources are allocated.

A.4 The Five Portfolios

Portfolio A: Nuclear Ceiling 40%

Nuclear 40%, Large Hydro 10%, Wind 15%, Solar 5%, CCGT 30%. Weighted output products: Nuclear 0.332, Hydro 0.037, Wind 0.051, Solar 0.013, CCGT 0.144. $\Sigma(x_i \times CF_i \times AF_i) = 0.577$. Carbon-weighted sum = 76.60. $G(A) = 76.60 / 0.577 = 132.8$ gCO₂/kWh.

Portfolio B: Nuclear Ceiling 30%

Nuclear 30%, Large Hydro 10%, Wind 15%, Solar 5%, CCGT 40%. $\Sigma(x_i \times CF_i \times AF_i) = 0.542$. Carbon-weighted sum = 99.13. $G(B) = 99.13 / 0.542 = 182.9$ gCO₂/kWh. Change from A: +50.1 gCO₂/kWh (+37.7%).

Portfolio C: Nuclear Ceiling 20%

Nuclear 20%, Large Hydro 10%, Wind 15%, Solar 5%, CCGT 50%. $\Sigma(x_i \times CF_i \times AF_i) = 0.507$. Carbon-weighted sum = 121.65. $G(C) = 121.65 / 0.507 = 239.9$ gCO₂/kWh. Change from B: +57.0 gCO₂/kWh (+31.2%).

Portfolio D: Nuclear Ceiling 10%

Nuclear 10%, Large Hydro 10%, Wind 15%, Solar 5%, CCGT 60%. $\Sigma(x_i \times CF_i \times AF_i) = 0.472$. Carbon-weighted sum = 144.18. $G(D) = 144.18 / 0.472 = 305.5$ gCO₂/kWh. Change from C: +65.6 gCO₂/kWh (+27.3%).

Portfolio E: Nuclear Ceiling 0% — Nuclear Excluded

Nuclear 0%, Large Hydro 10%, Wind 20%, Solar 10%, CCGT 60%. $\Sigma(x_i \times CF_i \times AF_i) = 0.418$. Carbon-weighted sum = 143.96. $G(E) = 143.96 / 0.418 = 344.4$ gCO₂/kWh. Change from D: +38.9 gCO₂/kWh (+12.7%).

A.5 The Sensitivity Summary

Table A.1: Portfolio Carbon Intensity as a Function of Nuclear Allocation Ceiling

Portfolio	Nuclear Allocation	CCGT Required	G(x) gCO ₂ /kWh	Change from Prior	
A	40%	30%	132.8	—	
B	30%	40%	182.9	+50.1 (+37.7%)	
C	20%	50%	239.9	+57.0 (+31.2%)	
D	10%	60%	305.5	+65.6 (+27.3%)	
E	0%	60%	344.4	+38.9 (+12.7%)	

A.6 Observations

Three observations follow directly from Table A.1.

First, the relationship between nuclear allocation and portfolio carbon intensity is monotonically increasing as nuclear is removed — every reduction worsens $G(x)$. The direction of the sensitivity result from Section 3.5 is confirmed numerically across the full range from 40% to 0%.

Second, the mechanism is CCGT substitution. In every portfolio, the demand constraint forces CCGT into the gap that reduced nuclear allocation creates, because no combination of the remaining low-carbon sources has sufficient $CF \times AF$ to satisfy the continuous load requirement without a dispatchable source.

Third, the deterioration is nonlinear. The carbon intensity penalty per percentage point of nuclear removed is not constant — it accelerates through the middle range and moderates slightly at the extremes. This reflects the compounding effect of CCGT substitution: as more CCGT enters the portfolio, its high carbon intensity dominates the weighted average with increasing leverage.

A.7 The Data Center Implication

The portfolios in this appendix use a demand constraint of $CF \times AF = 0.70$. A portfolio serving exclusively data center load would require $CF \times AF \geq 0.90$ — a substantially tighter constraint. At that demand target, the CCGT required to close the gap in each portfolio would be larger, the carbon intensity penalties more severe, and the nuclear allocation required to avoid significant CCGT penetration higher.

The numerical illustration therefore understates the carbon cost of nuclear exclusion for the specific application — serving continuous hyperscale data center load — that Section 4 identifies as the primary investment context. The sensitivity result from Section 3.5 is not only directionally confirmed by this appendix. For the load profile that motivates the paper, it is understated.